

Unit 1: Consumer Theory as a Modeling Framework

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1. Why consumer theory matters for research

- Consumer theory is not about describing behavior.
- It is a framework to model decision-making under constraints.
- A standard empirical paper in business economics typically:
 - assumes preferences
 - derives behavior
 - uses data to infer underlying parameters

Key idea: Preferences → Utility → Demand → Data

2. Preferences and their empirical content

- A preference relation $\succsim$ is assumed to be:
 - Complete
 - Transitive
 - Monotonic
 - Convex
- These assumptions impose restrictions on observable behavior.

Examples:

- Monotonicity → more is better → larger bundles should not be rejected
- Convexity → preference for diversification → mixtures preferred to extremes

These are not just theoretical assumptions but testable restrictions on behavior.

3. Utility representation

- If preferences satisfy regularity conditions, they can be represented by $u(x)$.
- Utility is not observable and is ordinal (only rankings matter).
- Empirical implication: preferences are identified up to monotonic transformations.

4. The consumer problem

- Max $u(x)$ subject to $P \cdot x \leq W$
- Primitives: preferences, prices, income
- Outcome: demand $x^*(P, W)$
- Demand is a mapping from primitives to observable behavior. This is the key step that allows us to move from theory to observable behavior.

5. Marshallian demand

- Solution to the utility maximization problem: $x^*(P, W)$.
- Depends on prices and income.
- Incorporates substitution and income effects.
- This is what we observe (indirectly) in data.

6. Indirect utility

- $v(P, W) = u(x^*(P, W))$
- Maximum utility attainable given prices and income.
- Properties:
 - Increasing in income
 - Decreasing in prices
 - Homogeneous of degree 0
- Only relative prices and income matter.

7. Expenditure minimization and duality

- Min $P \cdot x$ subject to $u(x) \geq \bar{u}$
- Solutions:
 - Hicksian demand $h(P, \bar{u})$
 - Expenditure function $e(P, \bar{u})$
- Interpretation:
 - Hicksian demand $\rightarrow$ cost-minimizing choices
 - Expenditure $\rightarrow$ minimum cost to reach utility level
- These tools allow us to evaluate welfare changes due to price or policy variations. This is what allows consumer theory to be used in policy and empirical evaluation.

8. Duality

- $v(P, e(P, \bar{u})) = \bar{u}$
- $e(P, v(P, W)) = W$
- Duality connects behavior and welfare and is central in empirical work.

9. From theory to data

- We do not observe utility or preferences. This is the fundamental identification problem in consumer theory.
- We observe choices, prices, and attributes.
- Key question: What patterns in observed choices are consistent with utility maximization?

10. Observable implications

- Models generate testable implications.
- Examples:
 - Higher prices $\rightarrow$ lower demand (ceteris paribus)
 - Trade-offs across attributes
 - Substitution across alternatives
- A key framework is “revealed preference theory”, which studies whether observed choices are consistent with utility maximization.
- WARP intuition: if a bundle is chosen when another was affordable, the latter should not be chosen in a situation where the first is also affordable.

11. Limitations

- Standard assumptions:
 - Stable preferences
 - Full information
 - Rational choice
- In reality:
 - Limited attention
 - Heuristics
 - Context dependence
- These limitations motivate modern extensions of the standard model.

12. Takeaway

Consumer theory is a discipline for thinking about decisions, not a literal description of reality. It provides structure, testable implications, and a bridge between theory and data.